SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical

PRACTICAL - 8 (M2)

AIM : Performing two-way ANOVA using aov() (R).

Input :

```
# S101 RAJDEEPP .M. PARAB
# PRACTICAL 8: TWO-WAY ANOVA
# Dataset : diabetes.csv
# Load the dataset
diabetes <- read.csv("diabetes.csv")
# View structure of dataset
str(diabetes)
# Create age groups
diabetes$age_group <- cut(diabetes$age,
                         breaks = c(0, 40, 60, 100),
                         labels = c("Young", "Middle", "Old"))
# Convert gender to factor
diabetes$gender <- as.factor(diabetes$gender)
# Perform two-way ANOVA
anova_two_way <- aov(glucose ~ gender * age_group,
                     data = diabetes)
# Display ANOVA table
summary(anova_two_way)
# HYPOTHESIS DECISION
p_values <- summary(anova_two_way)[[1]][["Pr(>F)"]]
if (any(p_values < 0.05, na.rm = TRUE)) {
  cat("Result: Reject the Null Hypothesis\n")
  cat("Reason: Gender, age group, or their interaction significantly affects glucose
levels.\n")
} else {
  cat("Result: Accept the Null Hypothesis\n")
  cat("Reason: No significant effect of gender or age group on glucose levels.\n")
}
```

Output :

```
> diabetes <- read.csv("C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 7 to 9 M2/diabetes.csv")
>   View(diabetes)
> # S101 RAJDEEPP .M. PARAB
> # PRACTICAL 8: TWO-WAY ANOVA
> # Dataset : diabetes.csv
> # Load the dataset
> diabetes <- read.csv("diabetes.csv")
> # View structure of dataset
> str(diabetes)
'data.frame':   10000 obs. of  20 variables:
 $ id               : int  1 2 3 4 5 6 7 8 9 10 ...
 $ age              : int  69 32 89 78 38 41 20 39 70 19 ...
 $ gender           : chr  "Male" "Male" "Female" "Male" ...
 $ bmi              : num  23.3 25 28.6 15.8 35.7 ...
 $ glucose          : int  170 184 87 96 171 98 98 171 160 74 ...
 $ blood_pressure   : int  137 177 164 113 122 107 135 126 114 157 ...
 $ cholesterol      : int  139 250 225 158 193 209 257 287 260 163 ...
 $ heart_rate       : int  66 75 102 112 109 60 89 93 113 93 ...
 $ sleep_hours      : num  6.4 6.4 7.5 6.9 7.4 9.9 8.7 5 6.7 4.6 ...
 $ physical_activity: chr  "High" "Medium" "Medium" "Low" ...
 $ smoking          : int  1 0 1 0 0 1 0 0 1 0 ...
 $ alcohol_intake   : int  1 1 0 1 1 0 0 0 1 1 ...
 $ family_history   : int  1 1 0 1 0 1 0 0 1 0 ...
 $ stress_level     : int  8 4 6 2 6 2 1 5 4 4 ...
 $ diet_score       : int  3 7 5 7 9 10 1 6 9 1 ...
 $ steps_per_day    : int  8760 7682 15025 4645 7862 4849 11604 7398 3735 16432 ...
 $ work_hours       : int  6 11 8 4 13 12 8 9 5 13 ...
 $ water_intake_ltr : num  3.6 4.1 1.4 4.4 3.9 3.1 4.7 4.1 3 4.5 ...
 $ insulin          : int  33 36 46 171 235 30 26 96 21 231 ...
 $ diabetes         : int  0 0 1 1 0 0 0 0 0 0 ...
> # Create age groups
> diabetes$age_group <- cut(diabetes$age,
+                           breaks = c(0, 40, 60, 100),
+                           labels = c("Young", "Middle", "Old"))
> # Convert gender to factor
> diabetes$gender <- as.factor(diabetes$gender)
```

```
 $ work_hours       : int  6 11 8 4 13 12 8 9 5 13 ...
 $ water_intake_ltr : num  3.6 4.1 1.4 4.4 3.9 3.1 4.7 4.1 3 4.5 ...
 $ insulin          : int  33 36 46 171 235 30 26 96 21 231 ...
 $ diabetes         : int  0 0 1 1 0 0 0 0 0 0 ...
> # Create age groups
> diabetes$age_group <- cut(diabetes$age,
+                           breaks = c(0, 40, 60, 100),
+                           labels = c("Young", "Middle", "Old"))
> # Convert gender to factor
> diabetes$gender <- as.factor(diabetes$gender)
> # Perform two-way ANOVA
> anova_two_way <- aov(glucose ~ gender * age_group,
+                      data = diabetes)
> # Display ANOVA table
> summary(anova_two_way)
                   Df   Sum Sq Mean Sq F value Pr(>F)
gender              1     2307  2306.6   1.627  0.202
age_group           2      586   292.8   0.207  0.813
gender:age_group    2     1924   961.8   0.678  0.507
Residuals        9994 14167899  1417.6
> # HYPOTHESIS DECISION
> p_values <- summary(anova_two_way)[[1]][["Pr(>F)"]]
> if (any(p_values < 0.05, na.rm = TRUE)) {
+   cat("Result: Reject the Null Hypothesis\n")
+   cat("Reason: Gender, age group, or their interaction significantly affects glucose levels.\n")
+ } else {
+   cat("Result: Accept the Null Hypothesis\n")
+   cat("Reason: No significant effect of gender or age group on glucose levels.\n")
+ }
Result: Accept the Null Hypothesis
Reason: No significant effect of gender or age group on glucose levels.
>
>
>
>
```